Paper I — External Adversarial Audit — Final Report

External Adversarial Audit — Paper I (Split-Graph Fractional Triangle-Packing Theorem)

Audit ID: PAPER_I_EXT_AUDIT_2026-07-10

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Auditor: independent external AI adversarial auditor (adversarial-math-audit protocol)

Request spec: `PAPER_I_EXTERNAL_FULL_AUDIT_REQUEST_SPEC.md`

Target (only): Theorem 1.1 of *Affine Profile Reduction for Fractional Triangle Packings in Split Graphs* —

> For every split graph G on n vertices, $|E(G)| - 2\cdot\nu\blacksquare^*(G) \le n^2/6 + n$,
> where $\nu\blacksquare^*(G)$ is the maximum fractional triangle-packing value.

Ground truth: `01_manuscript/PAPER_I_preprint_v1.0.{md,pdf,tex}` (Theorem 1.1, Appendix C); calculation ledger `02_validation/CALCULATION_LEDGER_…v1.0.md` (L1–L63); Lean sources `05_formalization/lean/`.

Prior internal status: ledger `FINAL_PASS / FROZEN / CLEAN`; manuscript claims the theorem is machine-verified in Lean, sorry-free, axioms `[propext, Classical.choice, Quot.sound]`.

1. Executive verdict

Verdict: PASS (intermediate AI-assurance tier — not human peer review)

The finite fractional split-graph theorem $|E(G)| - 2\nu\blacksquare^*(G) \le n^2/6 + n$ is **sound and correctly established** at every step the auditor could check independently, and the manuscript's scope is stated honestly (no over-claim into integral packing, clique partitions, chordal graphs, or Erd■s #81). Specifically:

- **Analytic proof (Gates A–G):** the two-phase packing, the exact deficit identity, the affine/concavity reduction to pure profiles, the orbit symmetrization, the three-orbit LP, the quadratic benchmark $R(p,q)$, and the conversion to n all reproduce exactly. Every closed-form algebraic identity of the ledger was re-derived **symbolically and found exact** (Script 01, 9/9). The residual-cover LP optimum equals the closed form $\min\{U,D,H\}$ and the full $C(p,2)$-variable LP equals the 3-orbit LP (orbit symmetrization lossless), and the affine reduction with uniform bound $R(p,q) - p/2$

holds on 400 random mixed profiles with **no counterexample** (Script 02).

- **Falsification (Gate F / kill-switch):** on 622 split graphs (structured + random + adversarial) the theorem was **never violated**; the extremal family K_p ∨ K■_{2p} attains n²/6 + n/6 exactly, confirming the leading constant 1/6 is tight and the +n term is a non-optimized slack (Script 03).
- **Formal certificate (Gate H): PASS.** An **independent rebuild** in a fresh source tree (no inherited `.lake` cache; community Mathlib cache) reports `Build completed successfully (8027 jobs), exit 0, with PaperI.paperI_main` **and** the LP-duality node `PaperI.Split.residual_duality` each depending on axioms `[propext, Classical.choice, Quot.sound] — no sorryAx, no native_decide`, no project-specific axiom. The released Lean sources match the recorded integrity manifest by SHA-256 (8/8), the Lean statement `paperI_main : Phi ≤ n²/6 + n` (`Phi = edgeCount − 2·nu3star`) is exactly Theorem 1.1, and a static scan finds no `sorry`/`admit`/`native_decide`/added `axiom` in the sources. This independently reproduces the recorded gate log.
- **Citations (Gate I):** the single proof-critical import — finite-dimensional LP strong duality — is standard, and the Lean development derives it from Mathlib's finite Farkas (`residual_duality`), so the result does not depend on the exact Schrijver reference.

No counterexample, circularity, unmet hypothesis, or scope over-claim was found. This is an independent AI adversarial audit: an intermediate assurance tier, stronger than self-audit but **not** external human peer review; it confers no publication status.

2. Scope, method, independence

In scope: Theorem 1.1 (finite fractional bound for split graphs) and its full proof chain, plus the Lean certificate for `paperI_main`.
Out of scope (confirmed absent from the claim): integral triangle packing, clique partitions/cp, chordal graphs, Φτ, asymptotic Erd■s #81, any fractional-to-integral transfer. Paper II and the asymptotic #81 route were not consulted.

Method: clean-context decomposition into claim IDs mapped to gates A–J; predeclared kill-switches for every computation; independent re-implementation of all primitives (ν■* LP, residual-cover LP, orbit LP, symbolic algebra); literal statement/definition cross-check across manuscript ↔ ledger ↔ Lean; independent Lean build in a fresh tree; SHA-256 provenance. **Independence:** fresh auditor code; fresh Lean source copy with no inherited `.lake` cache; the author's repo of record was not used. See `ENVIRONMENT.md`.

3. Gate table (A–J)

Gate	Item	Status	Evidence
A	Definitions (ν■*, deficit, split model, κ, profiles, orbits) well-posed & consistent across manuscript/ledger/Lean	**VERIFIED_WITHIN_SCOPE**	manuscript §2 ↔ ledger L1–L4 ↔ Lean `Split`,`Phi`,`nu3star`,`edgeCount`,n cross-checked
B	Aggregate-load / affine profile reduction; candidates suffice	**VERIFIED_WITHIN_SCOPE**	Lemma 3.1 (L4) exact; concavity L33/L34; Script 02(B) 400 mixed profiles, 0 counterexample
C	Orbit symmetrization + finite LP; solved independently	**VERIFIED_WITHIN_SCOPE**	Script 02: full C(p,2)-LP = 3-orbit LP = `min{U,D,H}` (0/660 & 0/280, err 5e-15)
D	Finite LP duality applied within hypotheses; no infinite-dim appeal	**VERIFIED_WITHIN_SCOPE**	App B: finite feasible bounded pair; weak duality + attainment; Lean derives from Farkas
E	Deficit identity (4.9/L28) + quantitative bound `R(p,q)`; constant recomputed	**VERIFIED_WITHIN_SCOPE**	Script 01: L28, L48, L50, L51, L53, L58, q=0 all exact (9/9)
F	Conversion to n (≤ $n^2/6+n$) + comparison with optimum packing; tightness	**VERIFIED_WITHIN_SCOPE**	L58–L63 exact; Script 03: 0 violations/622, extremal attains $n^2/6+n/6$
G	Assembly composes to the theorem; no circularity	**VERIFIED_WITHIN_SCOPE**	linear chain L1→L63; Lean `paperI_main` assembles `nu3star_ge_Vcom`+`Mcov_lower`+`edgeCount_eq`+arithmetic
H	Lean: build, sorry-freeness, `#print axioms`, statement match, SHA	**VERIFIED_WITHIN_SCOPE**	Script 04: independent rebuild 8027 jobs exit 0; `paperI_main`+`residual_duality` axioms `[propext,Classical.choice,Quot.sound]`, no sorryAx; SHA 8/8 match; statement = Thm 1.1
I	Citations (LP duality; split-graph structure) applied within hypotheses	**VERIFIED_WITH_RESIDUAL_RISK**	finite LP duality standard + Lean-derived; exact "Cor. 7.1g" label not book-checked
J	Over-claim / scope check (no chordal, cp, #81, integral)	**VERIFIED_WITHIN_SCOPE**	manuscript §9, ledger §15 explicitly disclaim; statement is fractional split-graph only

4. Independent computation (Scripts 01–03)

All scripts declared kill-switches before running; seed `20260710`; artifacts (script, `results.txt`, `report.pdf`, `.zip`, `.sha256`) under `scripts/`.

- **01 — algebraic identities (Gate E).** sympy exact check of L48, L50, L51 (SOS), L53, L58, the q=0 facts, and L25/L28. **9/9 exact** → `CALCULATION_VERIFIED`.
- **02 — orbit LP / symmetrization / reduction (Gates B, C).** (C0) q=0 branch verified; (C1) orbit LP = `min{U,D,H}` (0/660); (C2) full LP = orbit LP, err 5e-15 (0/280) — orbit averaging is lossless; (B) `M(κ) ≥ min_s M(κ^S)` and `M(κ) ≥ R−p/2` on 400 mixed profiles, 0 violations, bound tight (slack 0). Auditor correction CORR-02-01 recorded

(a q=0 domain over-reach in the first run; not a paper defect).

- **03 — split-graph falsification (Gate F).** 622 split graphs, **0** theorem

violations; max ratio $(|E|-2\nu■*)/(n^2/6+n) = 0.815$; extremal $K\_p \vee K■\_\{2p\}$ attains $n^2/6+n/6$ exactly (0 tightness failures).

5. Falsification / kill-switch log

No counterexample was found to Theorem 1.1 or to any gate. Failed falsification attempts (supporting evidence, not proof): adversarial dense complete-split graphs and 600 random split graphs (Script 03); random mixed neighborhood profiles probing the affine reduction and the uniform bound (Script 02); a search for an orbit/candidate beating `min{U,D,H}` (Script 02, none). The one issue surfaced was an **auditor** domain error (CORR-02-01), not a defect in the paper. Bounded search is existential evidence; the universal statement rests on the analytic proof + the Lean certificate.

6. Coverage declaration

Verified (within scope): all algebraic identities of the ledger (exact); the orbit LP closed form and the losslessness of orbit symmetrization (LP, bounded p); the affine reduction and uniform bound (LP, bounded p, random mixed profiles); the end-to-end inequality and extremal tightness (bounded split graphs); definition consistency across manuscript/ledger/Lean; Lean statement = Theorem 1.1; source SHA-256 = recorded manifest; static sorry/axiom-cleanliness of the sources; scope non-over-claim.

Residual risks: (i) numeric LP/enumeration checks are over bounded ranges ($p \le 11$ for the LP identities, $p \le 7–8$ for exact $\nu■*$), so they corroborate rather than prove the universal statement — but the analytic proof (Gates A–G) and the Lean certificate (Gate H) carry the universal claim; (ii) the exact wording of Schrijver "Corollary 7.1g" was not verified against the physical book (immaterial: the duality is standard and Lean-derived from Farkas); (iii) this is an AI audit, not human peer review.

7. Recommendation

Maintain the theorem at its internal `FINAL_PASS / FROZEN` status for the **finite fractional split-graph bound only**. The independent audit found the analytic proof, the computational checks, and the formal certificate mutually consistent and free of any detected error, circularity, or over-claim. Human expert / journal peer review remains the appropriate next tier; this AI audit confers no publication status.

Distinctions maintained: "analytic proof correct" (Gates A–G, I), "Lean certificate holds" (Gate H), and "computation found no counterexample" (Scripts 02–03) are reported separately; all three point the same way for Paper I.